MATH1025B
Essential Mathematical Methods

Chapter 9: Vectors in $\mathbb{R}^n$

November 13, 2025

$\mathbb{R}^n$ and Vector Operations

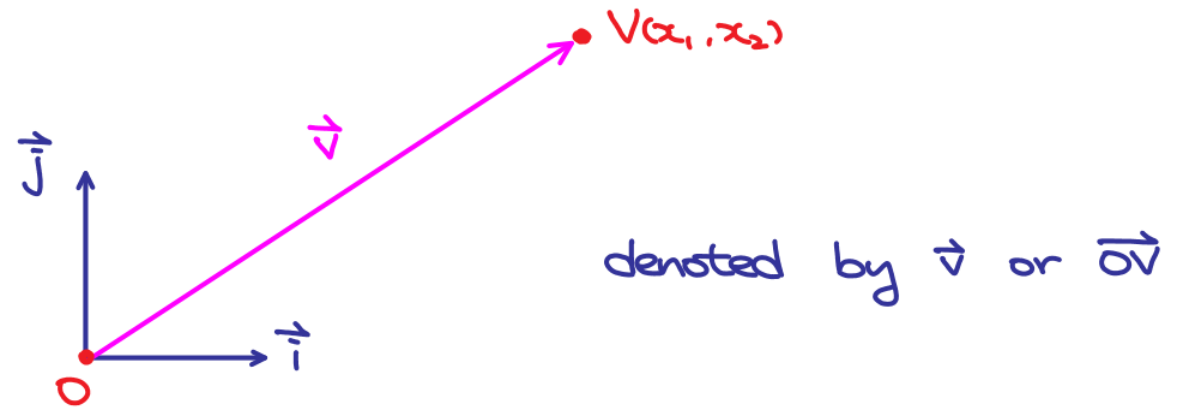
Definition 9.1

A **vector** in $\mathbb{R}^n$ is an element of

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{\text{n copies}} = \{(x_1, x_2, \ldots, x_n) : x_1, x_2, \ldots, x_n \in \mathbb{R}\}.$$

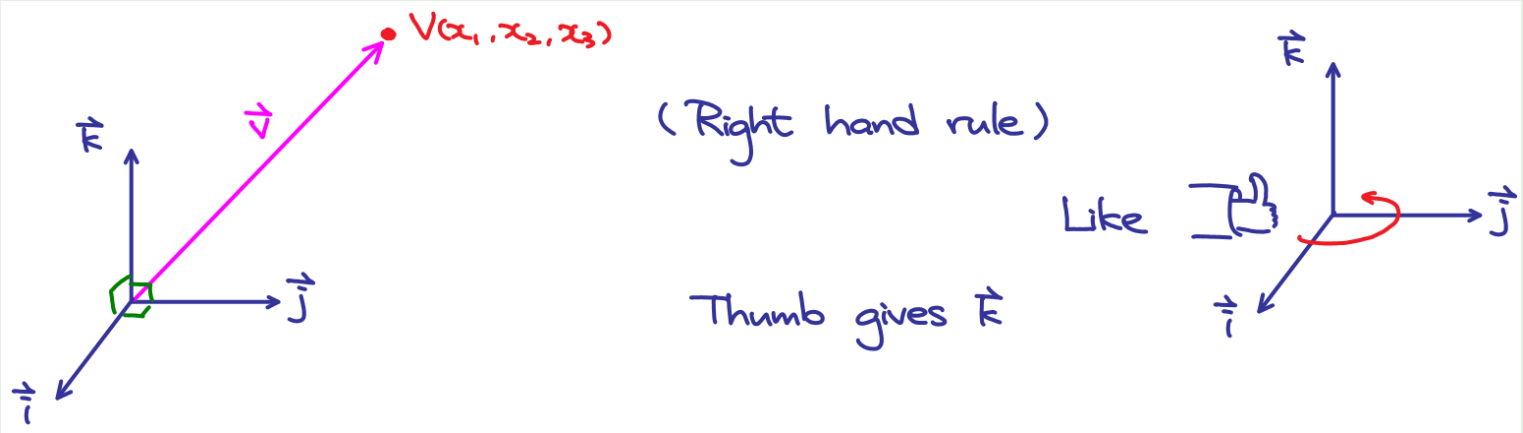
Example 9.1

- A vector in $\mathbb{R}^2$ can be written as (x_1, x_2) or $x_1\vec{i} + x_2\vec{j}$.



Example 9.1

- A vector in $\mathbb{R}^3$ can be written as (x_1, x_2, x_3) or $x_1\vec{i} + x_2\vec{j} + x_3\vec{k}$.



- A vector in $\mathbb{R}^n$ can be written as

$$(x_1, x_2, \ldots, x_n) \quad \text{or} \quad x_1\vec{e_1} + x_2\vec{e_2} + \cdots + x_n\vec{e_n},$$

where

$$\vec{e_i} = (0, \ldots, 0, \underset{\substack{\uparrow \\ i\text{-th}}}{1}, 0 \ldots, 0).$$

In particular, $\vec{0} = (0, 0, \ldots, 0) = 0\vec{e_1} + 0\vec{e_2} + \cdots + 0\vec{e_n}$ is said to be the zero vector.

$\mathbb{R}^n$ and Vector Operations

Definition 9.2 (Vector Addition)

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. Then

$$\vec{u} + \vec{v} := (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$

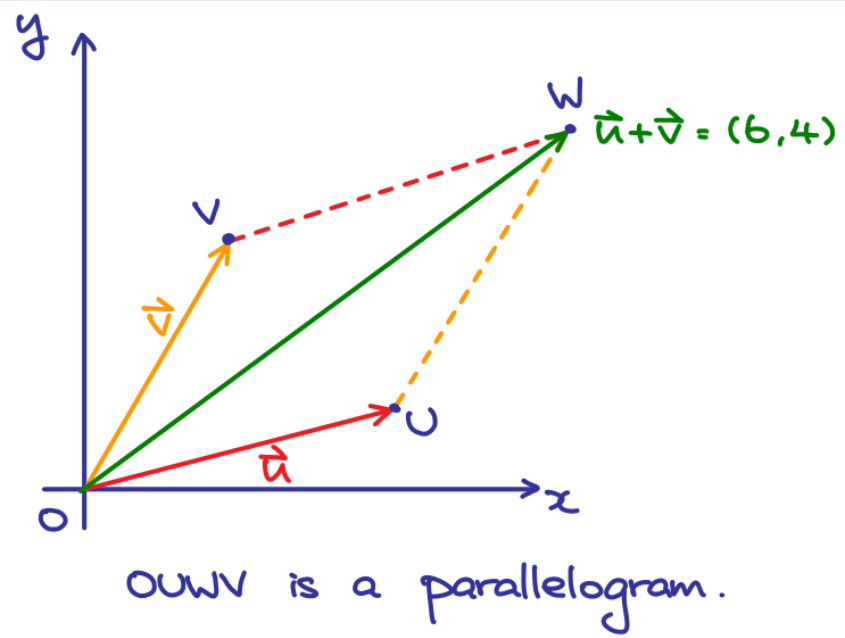
Example 9.2

Let $\overrightarrow{OU} = \vec{u} = 4\vec{i} + \vec{j}$, $\overrightarrow{OV} = \vec{v} = 2\vec{i} + 3\vec{j} \in \mathbb{R}^2$. Then

$$\vec{u} + \vec{v} = (4+2)\vec{i} + (1+3)\vec{j} = 6\vec{i} + 4\vec{j}.$$

We can also write

$$\overrightarrow{OU} + \overrightarrow{OV} = \overrightarrow{OU} + \overrightarrow{UW} = \overrightarrow{OW}.$$



Idea: Transport $\overrightarrow{OV} = \vec{v}$ so that its tail coincides with the tip of $\overrightarrow{OU} = \vec{u}$.

$\mathbb{R}^n$ and Vector Operations

Definition 9.2 (Vector Addition)

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. Then

$$\vec{u} + \vec{v} := (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$

Definition 9.3

Let $\vec{u}, \vec{v} \in \mathbb{R}^n$. Then

- $-\vec{v}$ is defined as the vector such that $\vec{v} + (-\vec{v}) = (-\vec{v}) + \vec{v} = \vec{0}$;
- $\vec{u} - \vec{v}$ is defined as $\vec{u} + (-\vec{v})$.

Example 9.3

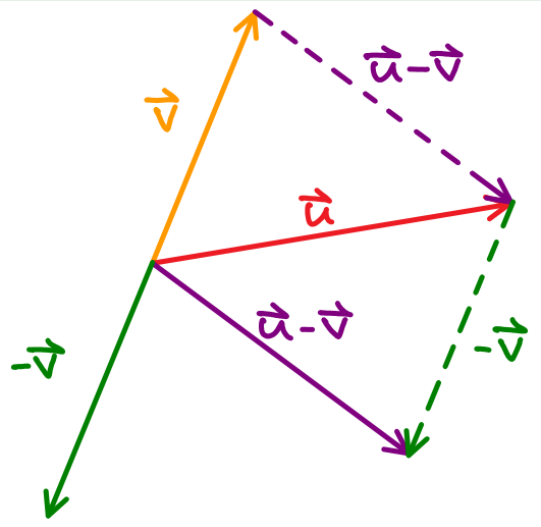
If $\overrightarrow{OU} = \vec{u} = 4\vec{i} + 2\vec{j}$ and $\overrightarrow{OV} = \vec{v} = \vec{i} + 3\vec{j}$, then

$$\overrightarrow{VO} = -\vec{v} = (-1)\vec{i} + (-3)\vec{j} \qquad (\text{Check: } \vec{v} + (-\vec{v}) = (-\vec{v}) + \vec{v} = \vec{0})$$

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v}) = (4\vec{i} + 2\vec{j}) + (-\vec{i} - 3\vec{j}) = 3\vec{i} - \vec{j}$$

We can also write:

$$\begin{aligned} \vec{u} - \vec{v} &= \overrightarrow{OU} - \overrightarrow{OV} = \overrightarrow{OU} + \overrightarrow{VO} \\ &= \overrightarrow{VO} + \overrightarrow{OU} = \overrightarrow{VU} \end{aligned}$$



Definition 9.4 (Scalar Multiplication)

If $\vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$, $t \in \mathbb{R}$ (called **scalar**), then

$$t\vec{v} := (tv_1, tv_2, \ldots, tv_n).$$

Remark: $-\vec{v} = (-1)\vec{v} = (-v_1, -v_2, \ldots, -v_n)$.

Definition 9.5

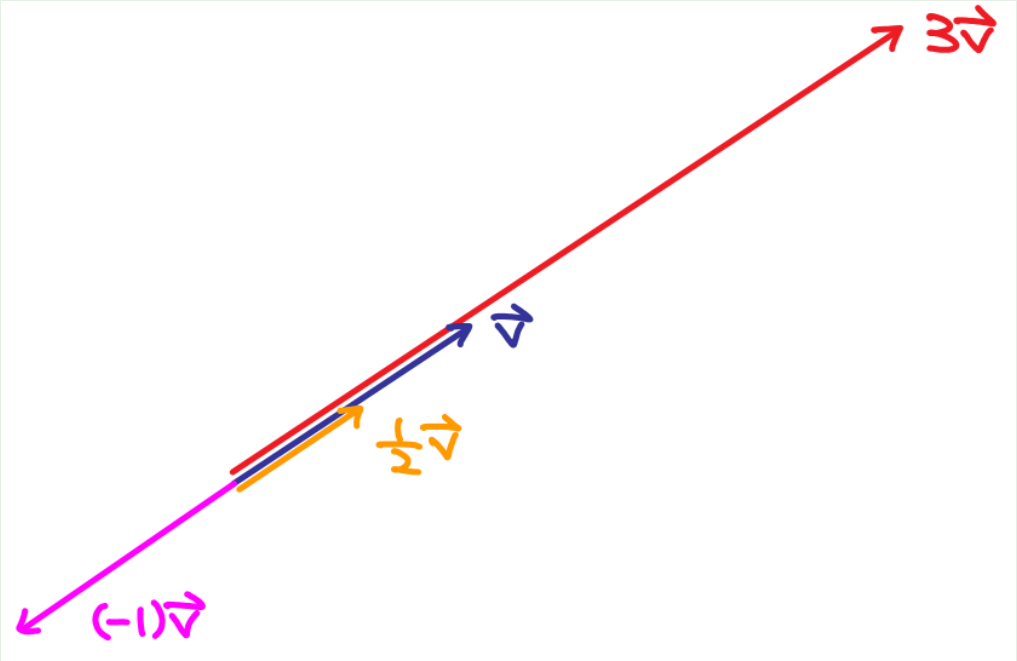
Two vectors are said to be **parallel** if one can be written as a scalar multiple of the other.

Remark: $\vec{0} \in \mathbb{R}^n$ is parallel to any vector $\vec{v} \in \mathbb{R}^n$ as $\vec{0} = 0\vec{v}$.

Example 9.4

If $\vec{v} = (4, 2) \in \mathbb{R}^n$, then

$$3\vec{v} = (12, 6), \quad \frac{1}{2}\vec{v} = (2, 1), \quad (-1)\vec{v} = (-4, -2).$$



Proposition 9.1

Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$, and let $s, t \in \mathbb{R}$.

(1) *(Commutative Law of Vector Addition)* $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

(2) *(Associative Law of Vector Addition)* $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$

(3) *(Existence of Additive Identity)* $\vec{0} + \vec{v} = \vec{v} + \vec{0} = \vec{v}$

(4) *(Existence of Additive Inverse)* $\vec{v} + (-\vec{v}) = (-\vec{v}) + \vec{v} = \vec{0}$

(5) *(Existence of Multiplicative Identity)* $1\vec{v} = \vec{v}$

(6) *(Associative Law of Scalar Multiplication)* $s(t\vec{v}) = (st)\vec{v}$

(7) *(Distributive Law of Scalar Multiplication)*

$$s(\vec{u} + \vec{v}) = s\vec{u} + s\vec{v} \quad \text{and} \quad (s + t)\vec{u} = s\vec{u} + t\vec{u}$$

Remark: $\mathbb{R}^n$ is a vector space.

Proof of (1):

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. Then

$$\begin{aligned}
\vec{u} + \vec{v} &= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n) \\
&= (v_1 + u_1, v_2 + u_2, \ldots, v_n + u_n) \quad (\because u_i, v_i \in \mathbb{R}, u_i + v_i = v_i + u_i) \\
&= \vec{v} + \vec{u}.
\end{aligned}$$

Definition 9.6

Let $\vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. **Length/Norm** of $\vec{v}$ is defined as

$$|\vec{v}| = \left(\sum_{i=1}^{n} v_i^2\right)^{1/2} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

(Or it is denoted by $\|\vec{v}\|$.)

Example 9.5

Let $\vec{v} \in \mathbb{R}^n$ and let $k \in \mathbb{R}$. Show that $|k\vec{v}| = |k||\vec{v}|$.

Let $\vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. Then

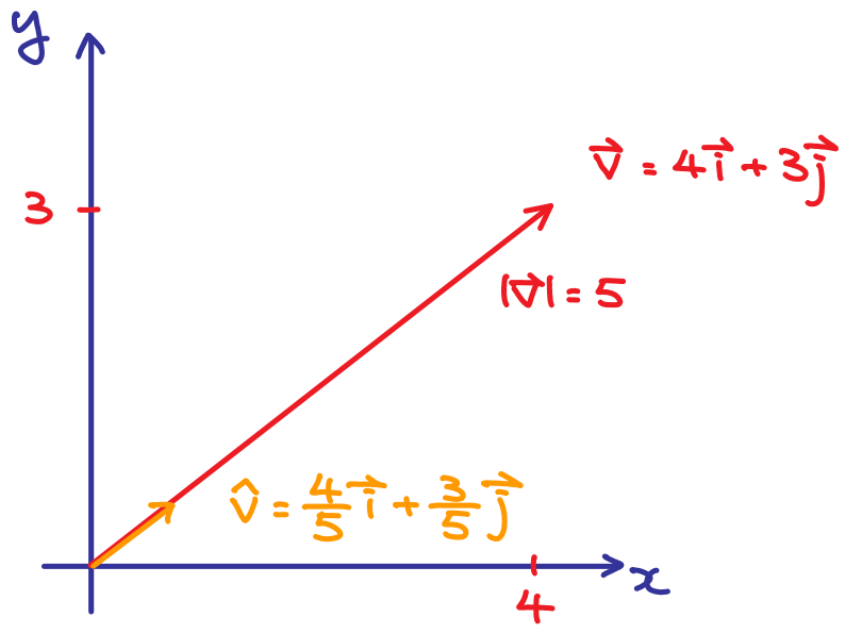
$$|k\vec{v}| = \left(\sum_{i=1}^{n} (kv_i)^2\right)^{1/2} = \left(k^2 \sum_{i=1}^{n} v_i^2\right)^{1/2} = |k||\vec{v}|.$$

(Note $|k| = \sqrt{k^2}$ for all $k \in \mathbb{R}$.)

If we let $\hat{v} = \dfrac{1}{|\vec{v}|}\vec{v}$, then $\hat{v} // \vec{v}$ and $|\hat{v}| = \left|\dfrac{1}{|\vec{v}|}\vec{v}\right| = \left|\dfrac{1}{|\vec{v}|}\right| |\vec{v}| = 1$.
Here $\hat{v}$ is said to be the unit vector of $\vec{v}$.
Idea: A vector $\vec{v}$ in $\mathbb{R}^n$ is a quantity with magnitude and direction. $\vec{v} = |\vec{v}|\hat{v}$ where $|\vec{v}|$ and $\hat{v}$ give the magnitude and direction respectively.

Example 9.6

Let $\vec{v} = 4\vec{i} + 3\vec{j} \in \mathbb{R}^2$.
Then $|\vec{v}| = \sqrt{4^2 + 3^2} = 5$ (Pyth. Thm.) and $\hat{v} = \frac{4}{5}\vec{i} + \frac{3}{5}\vec{j}$.



Dot Product

Definition 9.7 (Dot Product)

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. **The dot product of $\vec{u}$ and $\vec{v}$** is defined as

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

In particular, $\vec{v} \cdot \vec{v} = \sum_{i=1}^{n} v_i^2 = |\vec{v}|^2$.

Example 9.7

Let $\vec{u} = 4\vec{i} + 2\vec{j} + 3\vec{k}$, $\vec{v} = -2\vec{i} + 6\vec{j} + 5\vec{k}$. Then

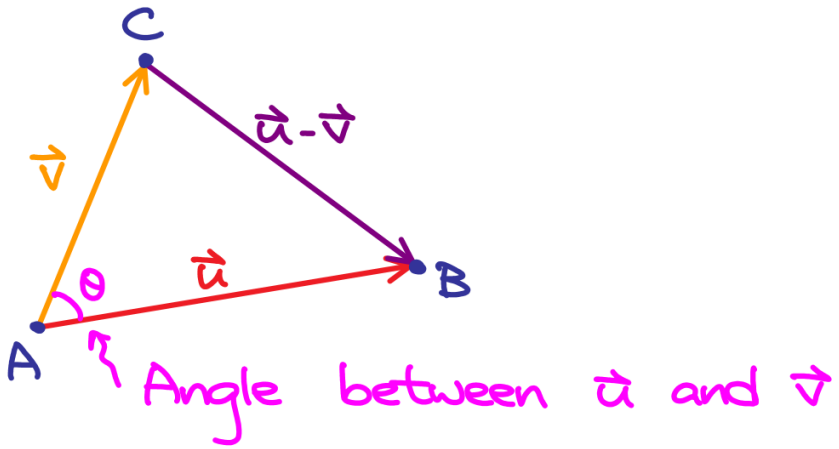
$$\vec{u} \cdot \vec{v} = 4 \cdot (-2) + 2 \cdot 6 + 3 \cdot 5 = 19,$$

and

$$\vec{v} \cdot \vec{u} = (-2) \cdot 4 + 6 \cdot 2 + 5 \cdot 3 = 19.$$

Well, what is the geometric meaning of $\vec{u} \cdot \vec{v}$?

Dot Product



Cosine Law: $BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos\angle BAC$

$$|\vec{u}-\vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$\sum_{i=1}^{n}(u_i - v_i)^2 = \sum_{i=1}^{n} u_i^2 + \sum_{i=1}^{n} v_i^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$\sum_{i=1}^{n} u_i^2 - 2\sum_{i=1}^{n} u_i v_i + \sum_{i=1}^{n} v_i^2 = \sum_{i=1}^{n} u_i^2 + \sum_{i=1}^{n} v_i^2 - 2|\vec{u}||\vec{v}|\cos\theta$$

$$\sum_{i=1}^{n} u_i v_i = |\vec{u}||\vec{v}|\cos\theta$$

So $\vec{u}\cdot\vec{v} = |\vec{u}||\vec{v}|\cos\theta$.

Definition 9.8 (Dot Product)

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$. **The dot product of $\vec{u}$ and $\vec{v}$** is defined as

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos\theta,$$

where θ $(0 \leq \theta \leq \pi)$ is the angle between $\vec{u}$ and $\vec{v}$ if $\vec{u}, \vec{v} \neq \vec{0}$.
Also, if either $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$, $\vec{u} \cdot \vec{v} = \vec{0}$.

Remark: Definition 9.7 defines the dot product in a computational way while definition 9.8 defines the dot product in a geometrical way, but actually the are equivalent.

Example 9.8

Draw $\vec{u} = 3\vec{i}$ and $\vec{v} = -2\vec{i} + 2\vec{j}$, and guess the angle between $\vec{u}$ and $\vec{v}$. Find the angle between $\vec{u}$ and $\vec{v}$ by using the formula $\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos\theta$.

$$\begin{aligned} \vec{u} \cdot \vec{v} &= |\vec{u}||\vec{v}| \cos\theta \\ -6 &= (3)(2\sqrt{2}) \cos\theta \\ \cos\theta &= -\frac{1}{\sqrt{2}} \\ \theta &= \frac{3\pi}{4} \end{aligned}$$

Dot Product

Direct consequences:

(1) $\vec{u}$ is perpendicular (or orthogonal) to $\vec{v}$, i.e. $\vec{u} \perp \vec{v}$, if and only if

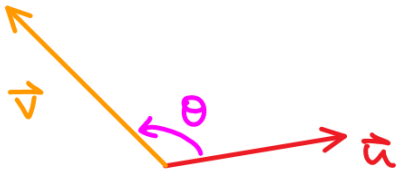
$$\theta = \frac{\pi}{2} \iff \vec{u} \cdot \vec{v} = 0.$$

(2)

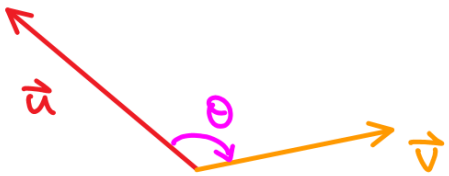
$$\vec{e}_i \cdot \vec{e}_j = \delta_{ij} = \begin{cases} |\vec{e}_i|^2 = 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

(δ_{ij} is called Kronecker delta)

Furthermore, let $\vec{u} = u_1\vec{i} + u_2\vec{j}, \vec{v} = v_1\vec{i} + v_2\vec{j} \in \mathbb{R}^2$.
Assume that θ is the angle measured from $\vec{u}$ to $\vec{v}$.

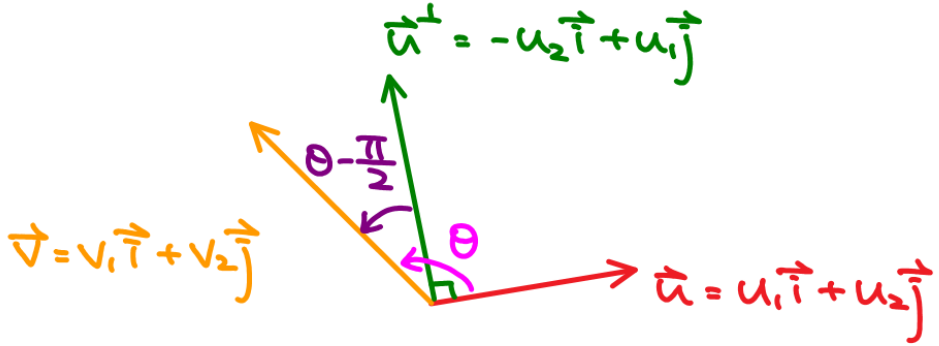


Anticlockwise direction, $\theta > 0$



Clockwise direction, $\theta < 0$.

Dot Product



The angle between $\vec{u}^{\perp}$ and $\vec{v}$ is $\theta - \frac{\pi}{2}$

The signed area of parllelogram spanned by $\vec{u}$ and $\vec{v}$

$$
\begin{aligned}
&= |\vec{u}||\vec{v}| \sin\theta \\
&= |\vec{u}||\vec{v}| \cos(\theta - \frac{\pi}{2}) \\
&= |\vec{u}^{\perp}||\vec{v}| \cos(\theta - \frac{\pi}{2}) \qquad (\because |\vec{u}| = |\vec{u}^{\perp}|) \\
&= \vec{u}^{\perp} \cdot \vec{v} \\
&= (-u_2\vec{i} + u_1\vec{j}) \cdot (v_1\vec{i} + v_2\vec{j}) \\
&= u_1v_2 - u_2v_1 = \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}.
\end{aligned}
$$

Exercise 9.1

Find the signed area of parallelogram spanned by $\vec{u}$ and $\vec{v}$ if

(1) $\vec{u} = \vec{i}, \vec{v} = \vec{j}$

(2) $\vec{u} = \vec{j}, \vec{v} = \vec{i}$

(3) $\vec{u} = 2\vec{i} + 3\vec{j}, \vec{v} = -4\vec{i} + \vec{j}$

Ans: (1) 1 (2) -1 (3) 14

Proposition 9.2

Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$, and let $t \in \mathbb{R}$.

(1) *(Commutative Law of Dot Product)* $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

(2) *(Distributive Law of Dot Product)* $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$

(3) *(Associative Law of Scalar and Dot Product)* $(t\vec{u}) \cdot \vec{v} = \vec{u} \cdot (t\vec{v}) = t(\vec{u} \cdot \vec{v})$

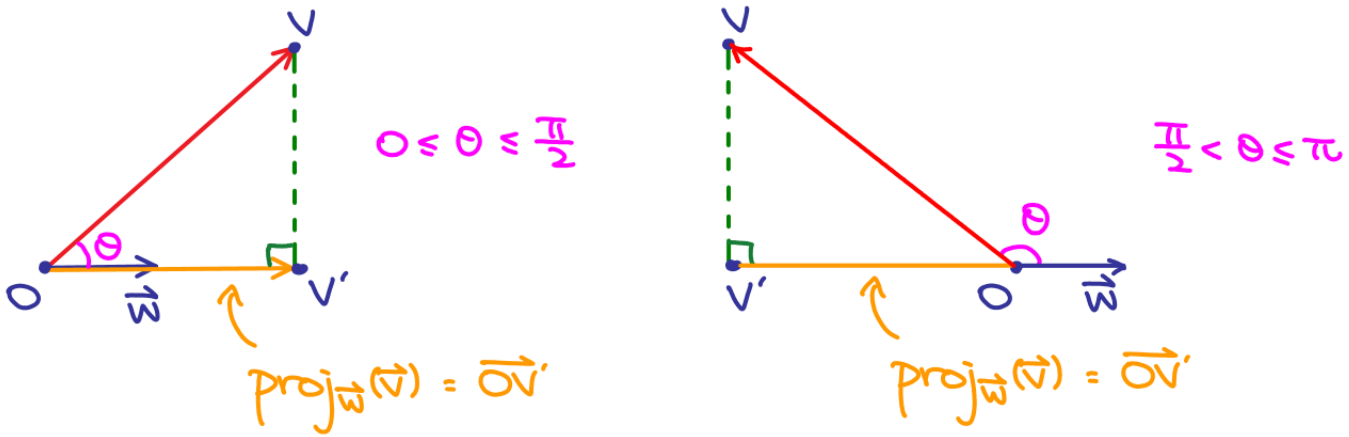
Proof of (2):
Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n)$ and $\vec{w} = (w_1, w_2, \ldots, w_n)$.

$$\begin{aligned}\vec{u} \cdot (\vec{v} + \vec{w}) &= \sum_{i=1}^{n} u_i(v_i + w_i) = \sum_{i=1}^{n} (u_i v_i + u_i w_i) \\ &= \sum_{i=1}^{n} u_i v_i + \sum_{i=1}^{n} u_i w_i = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}\end{aligned}$$

Furthermore,

$$(\vec{v} + \vec{w}) \cdot \vec{u} = \vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} = \vec{v} \cdot \vec{u} + \vec{w} \cdot \vec{u}.$$

Vector Projection



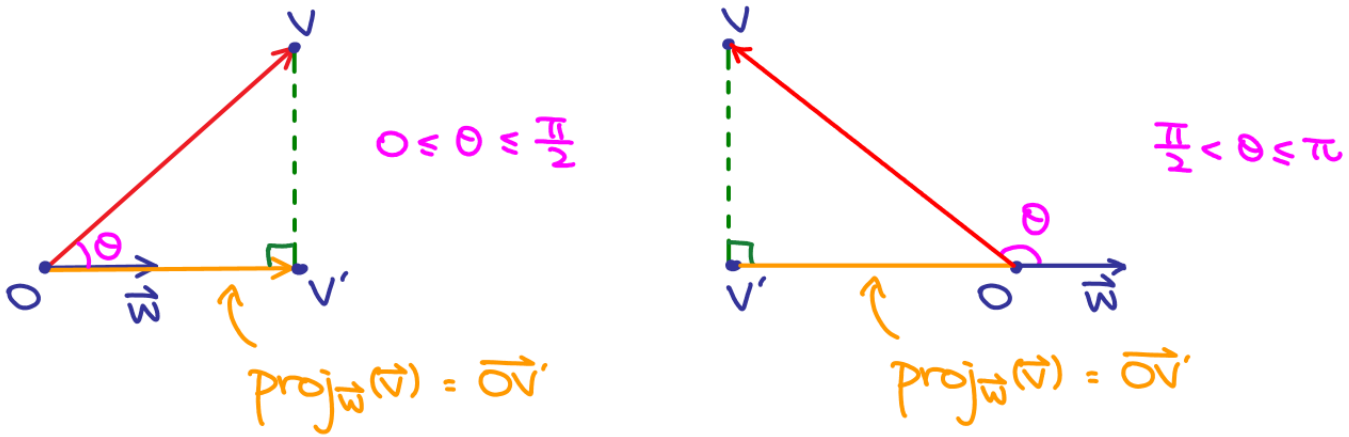
Projection of $\vec{v}$ along $\vec{w} = \overrightarrow{OV'}$:

	$0 \leq \theta \leq \frac{\pi}{2}$	$\frac{\pi}{2} \leq \theta \leq \pi$
Length of $\overrightarrow{OV'}$	$\lvert\vec{v}\rvert \cos\theta$	$-\lvert\vec{v}\rvert \cos\theta$
Direction of $\overrightarrow{OV'}$	$\hat{w}$	$-\hat{w}$

$$\text{proj}_{\vec{w}}(\vec{v}) = \overrightarrow{OV'} = (|\vec{v}|\cos\theta)\hat{w} = \left(\frac{|\vec{v}||\vec{w}|\cos\theta}{|\vec{w}|}\right)\left(\frac{\vec{w}}{|\vec{w}|}\right)$$

$$= \frac{\vec{v}\cdot\vec{w}}{|\vec{w}|^2}\vec{w}$$

Vector Projection



Furthermore, $\overrightarrow{OV} = \vec{v}$ can be expressed as

$$\overrightarrow{OV} = \overrightarrow{OV'} + \overrightarrow{V'V},$$

where

$$\overrightarrow{OV'} = \text{proj}_{\vec{w}}(\vec{v}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{w}|^2}\vec{w}$$

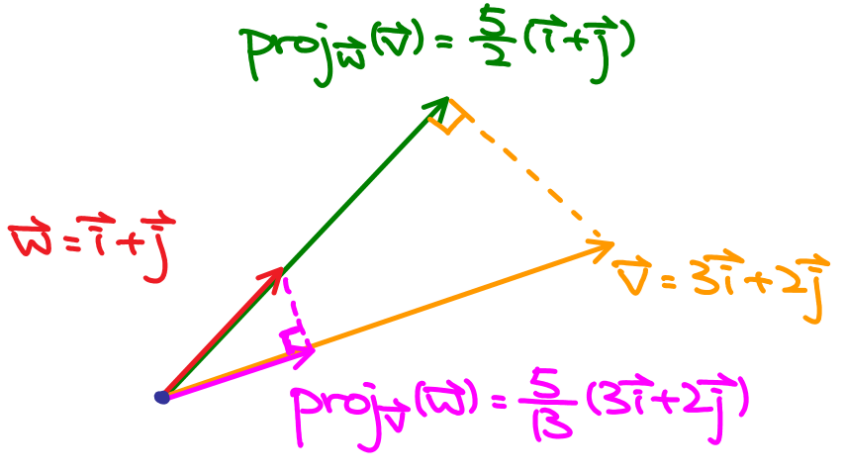
and

$$\overrightarrow{V'V} = \overrightarrow{OV} - \overrightarrow{OV'} = \vec{v} - \frac{\vec{v} \cdot \vec{w}}{|\vec{w}|^2}\vec{w}.$$

Then $\overrightarrow{OV'} // \vec{w}$ and $\overrightarrow{V'V} \cdot \vec{v} = (\vec{v} - \frac{\vec{v} \cdot \vec{w}}{|\vec{w}|^2}\vec{w}) \cdot \vec{w} = 0$, i.e. $\overrightarrow{V'V} \perp \vec{w}$.

Example 9.9

Let $\vec{v} = 3\vec{i} + 2\vec{j}, \vec{w} = \vec{i} + \vec{j} \in \mathbb{R}^2$.



Then

$$\text{proj}_{\vec{w}}(\vec{v}) = \frac{\vec{v}\cdot\vec{w}}{|\vec{w}|^2}\vec{w} = \frac{5}{2}(\vec{i}+\vec{j}),$$

$$\text{proj}_{\vec{v}}(\vec{w}) = \frac{\vec{w}\cdot\vec{v}}{|\vec{v}|^2}\vec{v} = \frac{5}{13}(3\vec{i}+2\vec{j}).$$

Exercise 9.2

Let $\vec{v} = 2\vec{i} + \vec{j} + 3\vec{k}, \vec{w} = \vec{i} - 2\vec{j} + 2\vec{k} \in \mathbb{R}^3$.
Express $\vec{v} = \vec{v}_1 + \vec{v}_2$ where $\vec{v}_1 // \vec{w}$ and $\vec{v}_2 \perp \vec{w}$.

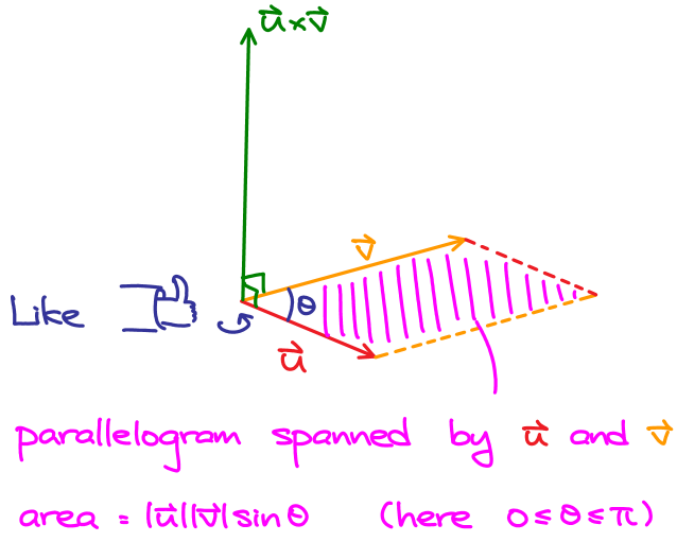
Ans: $\vec{v}_1 = \frac{2}{3}(\vec{i} - 2\vec{j} + 2\vec{k})$, $\vec{v}_2 = \frac{1}{3}(4\vec{i} + 7\vec{j} + 5\vec{k})$.

Cross Product in $\mathbb{R}^3$

Definition 9.9 (Cross Product in $\mathbb{R}^3$)

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. $\vec{u} \times \vec{v} \in \mathbb{R}^3$ is defined by the following:

- Magnitude: $|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}| \sin\theta$
- Direction: $\vec{u} \times \vec{v} \perp \vec{u}, \quad \vec{u} \times \vec{v} \perp \vec{v}$ with right hand rule



By definition, we have

(1) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$

(2) $\vec{i} \times \vec{j} = \vec{k}$, $\vec{j} \times \vec{k} = \vec{i}$, $\vec{k} \times \vec{i} = \vec{j}$
$\vec{i} \times \vec{i} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = \vec{0}$ (NOT the real number 0)

Cross Product in $\mathbb{R}^3$

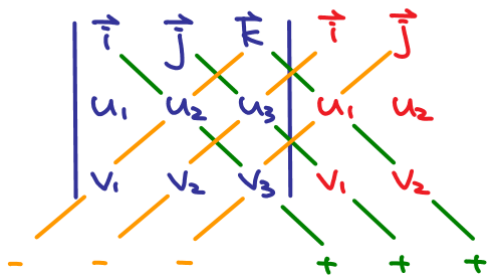
Question

How to compute $\vec{u} \times \vec{v}$ if $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$?

Assume linearity and distributive law (hard to show from definition 9.9):

$$
\begin{aligned}
\vec{u} \times \vec{v} &= (u_1\vec{i} + u_2\vec{j} + u_3\vec{k}) \times (v_1\vec{i} + v_2\vec{j} + v_3\vec{k}) \\
&= u_1v_1\vec{i} \times \vec{i} + u_1v_2\vec{i} \times \vec{j} + u_1v_3\vec{i} \times \vec{k} \\
&\quad u_2v_1\vec{j} \times \vec{i} + u_2v_2\vec{j} \times \vec{j} + u_2v_3\vec{j} \times \vec{k} \\
&\quad u_3v_1\vec{k} \times \vec{i} + u_3v_2\vec{k} \times \vec{j} + u_3v_3\vec{k} \times \vec{k} \\
&= (u_2v_3 - u_3v_2)\vec{i} - (u_1v_3 - u_3v_1)\vec{j} + (u_1v_2 - u_2v_1)\vec{k} \\
&= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}
\end{aligned}
$$

Sarrus Rule :



Cross Product in $\mathbb{R}^3$

Definition 9.10 (Cross Product in $\mathbb{R}^3$)

Let $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$. Then
$\vec{u} \times \vec{v} := (u_2v_3 - u_3v_2)\vec{i} - (u_1v_3 - u_3v_1)\vec{j} + (u_1v_2 - u_2v_1)\vec{k}$.

Remark: Definition 9.10 defines the dot product in a computational way while definition 9.9 defines the dot product in a geometrical way, but actually the are equivalent.

Example 9.10

Let $\vec{u} = \vec{i} + 2\vec{k}, \vec{v} = 2\vec{i} - 3\vec{j} + \vec{k}$. Then

$$\begin{aligned}
\vec{u} \times \vec{v} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2 \\ 2 & -3 & 1 \end{vmatrix} \\
&= \begin{vmatrix} 0 & 2 \\ -3 & 1 \end{vmatrix} \vec{i} - \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \vec{j} + \begin{vmatrix} 1 & 0 \\ 2 & -3 \end{vmatrix} \vec{k} \\
&= 6\vec{i} + 3\vec{j} - 3\vec{k}.
\end{aligned}$$

Cross Product in $\mathbb{R}^3$

Proposition 9.3

Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$, *and let* $t \in \mathbb{R}$.

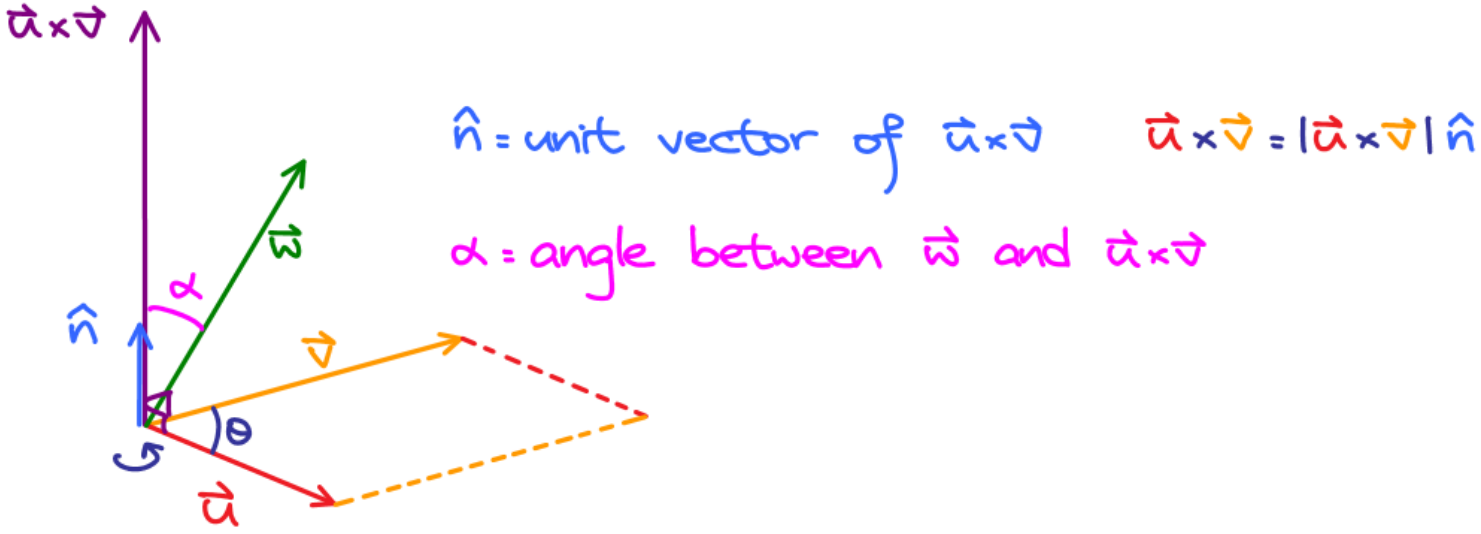
(1) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$

(2) *(Distributive Law of Cross Product)* $\quad \vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$

(3) $(t\vec{u}) \times \vec{v} = \vec{u} \times (t\vec{v}) = t(\vec{u} \times \vec{v})$

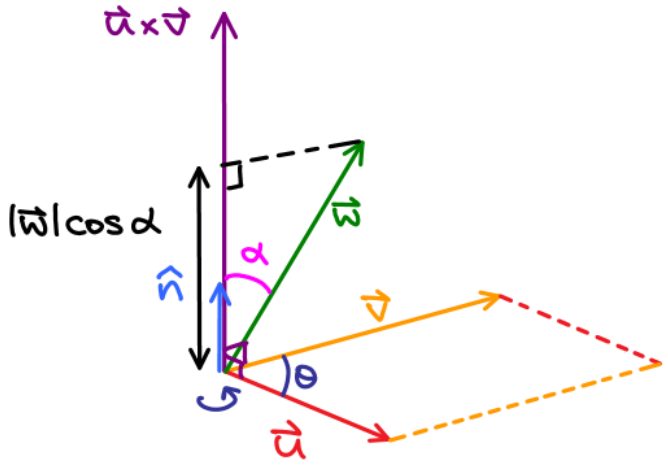
Scalar Triple Product

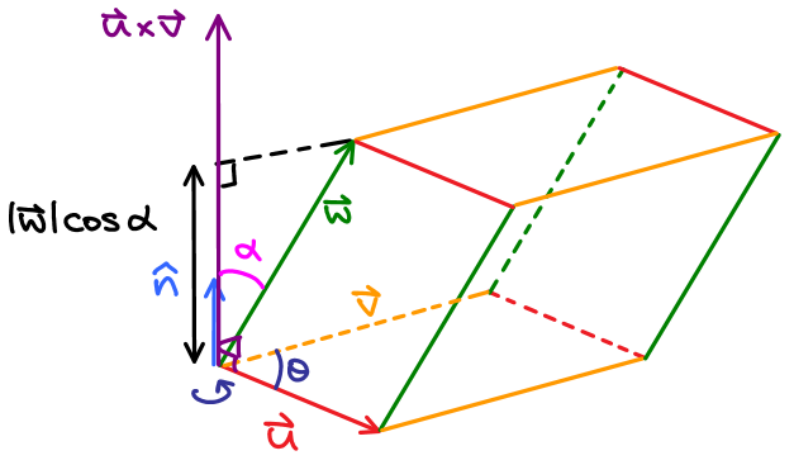
Note that if $\vec{u}, \vec{v} \in \mathbb{R}^3$, then $\vec{u} \times \vec{v} \in \mathbb{R}^3$.
Furthermore, suppose $\vec{w} \in \mathbb{R}^3$. Then, we know that $(\vec{u} \times \vec{v}) \cdot \vec{w}$ is well-defined and it is just a scalar.
Here $(\vec{u} \times \vec{v}) \cdot \vec{w}$ is called **scalar triple product**.
What is the geometrical meaning?



$$
\begin{aligned}
(\vec{u} \times \vec{v}) \cdot \vec{w} &= |\vec{u} \times \vec{v}| \hat{n} \cdot \vec{w} \\
&= |\vec{u} \times \vec{v}| (|\vec{w}| \cos \alpha)
\end{aligned}
$$

Scalar Triple Product





$$
\begin{aligned}
(\vec{u}\times\vec{v})\cdot\vec{w} &= |\vec{u}\times\vec{v}|\hat{n}\cdot\vec{w} \\
&= |\vec{u}\times\vec{v}|(|\vec{w}|\cos\alpha) \\
&= \underbrace{|\vec{u}\times\vec{v}|}_{\text{base area}}\ \underbrace{(|\vec{w}|\cos\alpha)}_{\text{height (with } \pm \text{ sign)}} \\
&= \text{signed volume of the paralelepiped spanned by } \vec{u}, \vec{v} \text{ and } \vec{w}
\end{aligned}
$$

(Every face of a parallelepiped is a parallelogram.)

Remark: If $\frac{\pi}{2} < \alpha \leq \pi$, $\cos\alpha < 0$ and so $(\vec{u}\times\vec{v})\cdot\vec{w} < 0$.

Scalar Triple Product

Let $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$, $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ and $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$. Then

$$
\begin{aligned}
&(\vec{u} \times \vec{v}) \cdot \vec{w} \\
&= \left[(u_2v_3 - u_3v_2)\vec{i} - (u_1v_3 - u_3v_1)\vec{j} + (u_1v_2 - u_2v_1)\vec{k}\right] \cdot (w_1\vec{i} + w_2\vec{j} + w_3\vec{k}) \\
&= (u_2v_3 - u_3v_2)w_1 - (u_1v_3 - u_3v_1)w_2 + (u_1v_2 - u_2v_1)w_3 \\
&= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}
\end{aligned}
$$

By properties of determinants:

$$
\begin{aligned}
(\vec{u} \times \vec{v}) \cdot \vec{w} &= (\vec{v} \times \vec{w}) \cdot \vec{u} = (\vec{w} \times \vec{u}) \cdot \vec{v} \\
(\vec{v} \times \vec{u}) \cdot \vec{w} &= (\vec{w} \times \vec{v}) \cdot \vec{u} = (\vec{u} \times \vec{w}) \cdot \vec{v}
\end{aligned}
$$

and the two rows differ by a minus sign.

Cauchy-Schwarz's Inequality

Let $\vec{u} = (u_1, u_2, \ldots, u_n), \vec{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$.
Dot product:

$$\vec{u} \cdot \vec{v} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

In particular, $\vec{v} \cdot \vec{v} = \sum_{i=1}^{n} v_i^2 = |\vec{v}|^2$.

Theorem 9.4 (Cauchy-Schwarz's Inequality)

Let $\vec{x} = (x_1, x_2, \ldots, x_n)$, $\vec{y} = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$. Then

$$|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \, \|\vec{y}\|.$$

Moreover, the equality holds

$$\begin{aligned} &\iff \text{there exists } \alpha \in \mathbb{R} \text{ such that } \vec{x} = \alpha \vec{y} \text{ or } \vec{y} = \alpha \vec{x} \\ &\iff \vec{x} \text{ and } \vec{y} \text{ are parallel.} \end{aligned}$$

Triangle Inequality

Theorem 9.5 (Triangle Inequality)

Let $\vec{x} = (x_1, x_2, \ldots, x_n)$, $\vec{y} = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$. *Then*

$$\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|.$$

Moreover, the equality holds if and only if

$$\vec{x} = \alpha\vec{y} \quad \textit{for some } \alpha \geq 0 \quad \textit{or} \quad \vec{y} = \alpha\vec{x} \quad \textit{for some } \alpha \geq 0.$$

Proof:

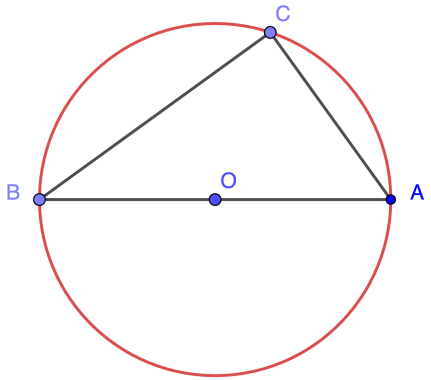
$$\begin{aligned}
\|\vec{x} + \vec{y}\|^2 &= (\vec{x} + \vec{y}) \cdot (\vec{x} + \vec{y}) \\
&= \vec{x} \cdot \vec{x} + \vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{x} + \vec{y} \cdot \vec{y} \\
&= \|\vec{x}\|^2 + 2(\vec{x} \cdot \vec{y}) + \|\vec{y}\|^2 \\
&\leq \|\vec{x}\|^2 + 2|\vec{x} \cdot \vec{y}| + \|\vec{y}\|^2 && (1) \\
&\leq \|\vec{x}\|^2 + 2\|\vec{x}\|\|\vec{y}\| + \|\vec{y}\|^2 && (2) \\
&= (\|\vec{x}\| + \|\vec{y}\|)^2.
\end{aligned}$$

The equalities hold in both (1) and (2) if and only if

$$\vec{x} = \alpha\vec{y} \quad \text{for some } \alpha \geq 0 \quad \text{or} \quad \vec{y} = \alpha\vec{x} \quad \text{for some } \alpha \geq 0.$$

Exercise 9.3

Consider a circle centered at O such that AB is its diameter. Let C be a point on the circle. Show that $\angle ACB = \frac{\pi}{2}$.



$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC}$$
$$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -\overrightarrow{AO} + \overrightarrow{OC}$$

$$\begin{aligned}
\overrightarrow{AC} \cdot \overrightarrow{BC} &= (\overrightarrow{AO} + \overrightarrow{OC}) \cdot (-\overrightarrow{AO} + \overrightarrow{OC}) \\
&= -\overrightarrow{AO} \cdot \overrightarrow{AO} + \overrightarrow{AO} \cdot \overrightarrow{OC} - \overrightarrow{OC} \cdot \overrightarrow{AO} + \overrightarrow{OC} \cdot \overrightarrow{OC} \\
&= -|\overrightarrow{AO}|^2 + |\overrightarrow{OC}|^2 \\
&= 0 \qquad (|\overrightarrow{AO}| = |\overrightarrow{OC}| \text{ are radius})
\end{aligned}$$

So $\overrightarrow{AC} \perp \overrightarrow{BC}$, i.e. $\angle ACB = \frac{\pi}{2}$.

Exercise 9.4

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Show that

$$2|\vec{x}|^2 + 2|\vec{y}|^2 = |\vec{x} + \vec{y}|^2 + |\vec{x} - \vec{y}|^2.$$

This is the parallelogram law:

the sum of the squares of the lengths of the four sides of a parallelogram equals the sum of the squares of the lengths of the two diagonals.

Cross Product and Scalar Triple Product

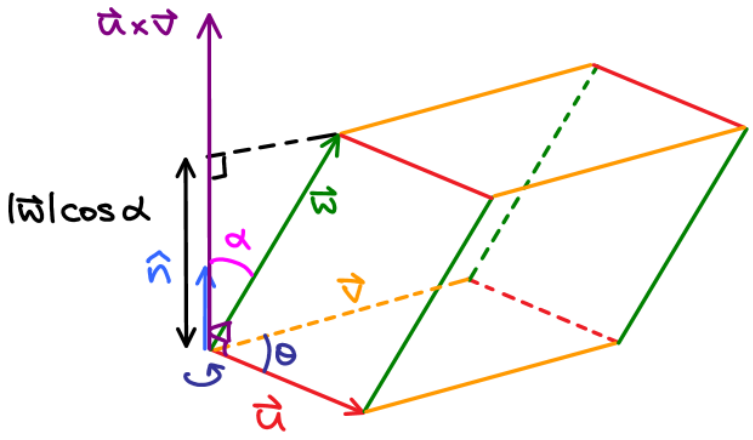
Let $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$, $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ and $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$.

Cross Product:

$$\begin{aligned}
\vec{u} \times \vec{v} &:= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \\
&= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \vec{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \vec{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \vec{k} \\
&= (u_2v_3 - u_3v_2)\vec{i} - (u_1v_3 - u_3v_1)\vec{j} + (u_1v_2 - u_2v_1)\vec{k}
\end{aligned}$$

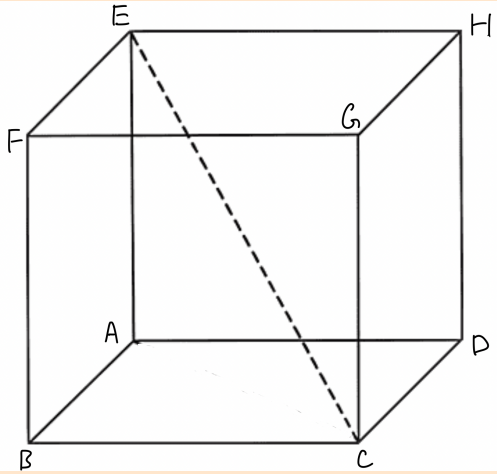
Scalar Triple Product:

$$\begin{aligned}
(\vec{u} \times \vec{v}) \cdot \vec{w} &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \\
&= \text{signed volume of the parallelepiped spanned by } \vec{u}, \vec{v} \text{ and } \vec{w}
\end{aligned}$$



Exercise 9.5

As shown in the figure is a cube with side length 2. Find $(\overrightarrow{AB} \times \overrightarrow{AD}) \cdot \overrightarrow{CE}$.



Ans: 8

Exercise 9.6

Let $\vec{a} = (1, 2, 3)$, $\vec{b} = (3, 2, 1)$ and $\vec{c} = (x, y, z)$. Suppose $|\vec{c}| = \sqrt{6}$. Find the maximum value of the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ x & y & z \end{vmatrix}$.

Ans: 24